

# Clustering Enhancement using Particle Swarm Optimization Algorithm

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**Abstract**—Clustering aims to find useful groups of objects (clusters), based on the closest distance to clusters centers. Soft Subspace Clustering algorithms are generally formulated as an optimization problem that minimizes an objective function, which makes a great improvement in classification. However, each time several iterations to determine the best cluster center are required. In addition, objective function optimization does not always find the global minimum but some local ones. In this paper, We have introduced the algorithm of Particle Swarm Optimization (PSO), which is combined with two subspace clustering algorithms to enhance the clusters centers positions. the proposed classification method has been tested on a synthetic and aerial data set. Two metrics have been used to compare and validate the final results: Normalized Mutual Information (NMI) and classification accuracy. The clustering results obtained are good and improved.

**Index Terms**—Clustering , soft subspace Clustering , objective function, Particle Swarm Optimization (PSO), cluster center.

## I. INTRODUCTION

In subspace clustering [1], clusters are extracted using a selected subset of features. Soft subspace clustering [2] is a type of subspace clustering, which assigns weights to different features in different clusters. These weights reflect their contributions to the corresponding cluster. This type has achieved great success in different domains, machine learning, and databases. It has known several improvements over the past ten years. Soft subspace clustering algorithms are based on the fundamental idea of optimization an objective function[3]. Various objective functions have been used in different soft subspace clustering algorithms; Among these methods we can mention, Entropy Weighting k-means algorithm (EWKM) [4] in this algorithm, the negative Shannon entropy is used to stimulate more feature dimensions to clusters identification; the the objective function of the Enhanced soft subspace clustering (ESSC) [5] algorithm was composed by the within-cluster compactness and the between-clusters separation. these two criteria are also used by Active Learning Soft Subspace clustering algorithm (ALISSC) with the clusters density, the support vector machine (SVM) [7] is combined with this

function for improving the clustering results.

Soft subspace clustering algorithms optimize the objective function by gradient-based methods, which can lose the local optimum, to overcome this problem a meta-heuristic methods showed. Different approaches have been proposed for optimizing clustering challenges; Genetic Algorithm (GA) [8] Ant Colony Optimization (ACO) [9,10], and Particle Swarm Optimization (PSO)[11]. PSO is the most popular Swarm Intelligence technique and one of the favorite areas of researchers. Several classification works used the PSO algorithm [12-13] to improve the classification results . For example a hybrid model is proposed in [12] that contains the Bisecting K-mean algorithm and Particle Swarm optimization algorithm. This method use the PSO to deal with the problem of random initialization of cluster centers and realize the optimum allowed path. In [13], PSO has been used to improve the FCM algorithm .

In this paper, PSO are used alternately with two objective functions of soft subspace clustering algorithms.

The structure of this paper is presented as follows. the idea of the proposed method has been explained in the section 2. Section 3, presents the experimental results of compared algorithms. In section 4, the conclusion was drawn.

## II. PROPOSED METHOD

A Particle Swarm Optimization (PSO) meta-heuristic is used to improve the clusters center positions. Two objective functions are used alone; (ALISSC[6] and ESSC[5]). PSO and an objective functions of soft subspace clustering algorithms are combined .This hybrid approach is used alternately; in the first iteration soft subspace clustering algorithms seeks solutions that minimize the objective function. These solutions are then used as an initialization step of the particles. In the general case of PSO, at each iteration, the position of a particle is updated according to its personal best position and the best solution of the swarm. The new solutions provided are in turn used to boost the soft subspace clustering.

The idea of this method was applied using the ESSC function for the first method and ALISSC for the second one.

### A. Objective function

The two objective functions of the enhanced soft subspace clustering algorithm (ESSC)[5] and (active learning Improving soft subspace clustering (ALISSC))[6] are written by the equations 1 and 2:

$$J_{ESSC}(v, u, w) = \sum_{i=1}^c \sum_{j=1}^N u_{ij}^m \sum_{k=1}^D w_{ik} (x_{jk} - v_{ik})^2 + \gamma \sum_{i=1}^c \sum_{k=1}^D w_{ik} \ln(w_{ik}) - \eta \sum_{i=1}^c \sum_{j=1}^N u_{ij}^m \sum_{k=1}^D w_{ik} (v_{ik} - v_{0k})^2 \quad (1)$$

$$J_{ALLISC}(v, u) = \frac{\sum_{i=1}^c u_{ij}^m \sum_{k=1}^D w_{ik} (x_{jk} - v_{ik})^2}{(n_{ik} + 1)} - \eta \sum_{i=1}^c \sum_{j=1}^N u_{ij}^m \sum_{k=1}^D w_{ik} (v_{ik} - v_{0k})^2 \quad (2)$$

With:

v: Cluster Centre  
w: weight matrix  
u: degree of membership  
C: clusters number  
N: Data size  
v<sub>0</sub>: Center of center  
m: fuzzy parameter  
D: Size of attributes (subspaces)  
x: Object / Gray Level / Color  
n<sub>ik</sub>: Density of the first class in the k th sub-space  
γ, η: Parameters affecting the value of the weights.  
Using the Lagrang multipliers, u, v and w minimize the objective function of EQ.1. are defined as:

$$u_{ij} = \frac{[\sum_{k=1}^D w_{ik} (x_{jk} - v_{ik}) - \eta \sum_{k=1}^D w_{ik} (v_{ik} - v_{0k})^2]^P}{[\sum_{i=1}^c \sum_{k=1}^D w_{ik} (x_{jk} - v_{ik}) - \eta \sum_{k=1}^D w_{ik} (v_{ik} - v_{0k})^2]^P} \quad (3)$$

with P=-1/(m-1)

$$v_{ik} = \frac{\sum_{j=1}^N u_{ij}^m (x_{jk} - v_{ik}) - \eta \sum_{j=1}^N u_{ij}^m (v_{ik} - v_{0k})^2}{\sum_{j=1}^N u_{ij}^m (1 - \eta)} \quad (4)$$

$$w_{jk} = \frac{\exp \frac{\sum_{j=1}^N u_{ij}^m (x_{jk} - v_{ik}) - \eta \sum_{j=1}^N u_{ij}^m (v_{ik} - v_{0k})^2}{\gamma}}{\sum_{k=1}^D \exp \frac{\sum_{j=1}^N u_{ij}^m (x_{jk} - v_{ik}) - \eta \sum_{j=1}^N u_{ij}^m (v_{ik} - v_{0k})^2}{\gamma}} \quad (5)$$

Using the Lagrang multipliers, u, v and w minimize the objective function of EQ.2 are defined as:

$$u_{ij} = \frac{[\sum_{k=1}^D w_{ik} (\frac{(x_{jk} - v_{ik})^2}{n_{ik} + 1} - \eta (v_{ik} - v_{0k})^2)]^{\frac{-1}{m-1}}}{\sum_{i=1}^c [\sum_{k=1}^D w_{ik} (\frac{(x_{jk} - v_{ik})^2}{n_{ik} + 1} - \eta (v_{ik} - v_{0k})^2)]^{\frac{-1}{m-1}}} \quad (6)$$

$$v_{ij} = \frac{\sum_{j=1}^N u_{ij}^m \frac{(x_{jk} - v_{ik})}{n_{ik} + 1} - \eta (v_{ik})}{\sum_{j=1}^N u_{ij}^m \frac{1}{n_{ik} + 1} - \eta} \quad (7)$$

$$w_{ik} = \frac{sm_{ik}}{\sum_{k=1}^D sm_{ik}} \quad (8)$$

with

$$sm_{ik} = \sum_{h=1}^{N_i} ent(h, k)$$

$$Underduress: \sum_{k=1}^D W_{ik} = 1$$

ent:entropy of data

N<sub>i</sub>: number of objects in a cluster i

### B. Particle Swarm Optimization

Particle Swarm Optimization (PSO) is an optimization method that allows to find the optimum of a function or fitness. This function (aptitude) defined by the user. PSO is based on the movement of individuals or particles in their search space, assigning them an initial position and velocity defined by the equations 9 and 10. The process will be repeated until these particles converge to a global minimum.

$$v_{ij}(t+1) = w * v_{ij}(t) + c_1 r_1 [Pbest_{ij}(t) x_{ij}(t)] + c_2 r_2 [Gbest_{ij}(t) x_{ij}(t)] \quad (9)$$

$$x_{ij}(t+1) = x_{ij}(t) + v_{ij}(t) \quad (10)$$

where

x:particle position. v: particle velocity.

w: inertial weight

Pbest: the best particle position.

Gbest: global best position of all swarms

c<sub>1</sub>, c<sub>2</sub>: two positive constants (coefficients/percent acceleration).

r<sub>1</sub>, r<sub>2</sub>: random values between 0 and 1.

The parameters used by PSO are given by the Table.1

Algorithm 1 shows proposed methods; the same algorithm with two different objective functions, ESSC and ALISSC

#### Algorithm 1

**step 1:** preparation of data:

- Data normalization.

- Initialization of clustering parameters v and w

- Calculate u used EQ.3 for first method and EQ.6 for second method.

- Initialization of PSO parameters

**Step2:** iterative processes

- started PSO for obtained the best cluster solution.

- Recalculated w used equation EQ.5 for first method and EQ.8 for second method

- Recalculated new degrees used best cluster with EQ.3 and EQ.6

- **repeat step 2 until convergence.**

**Step 3:**Classified data

The memberships degrees is used for a classification step

**End**

## III. RESULTS AND DISCUSSIONS

Different synthetic images and datasets used to test the two proposed methods.

### A. Synthetic datasets

A description of the datasets is given in Table.2. To evaluate the clustering results, the normalized mutual information (NMI) [15] and the classification accuracy are used. NMI is given by EQ.11.

$$NMI = \frac{\sum_{i=1}^c \sum_{j=1}^c [N_{ij} * \log((N * N_{ij}) / (N_j * N_i))]}{\sqrt{\sum_{i=1}^c N_i * \log(N_i / N) * \sum_{j=1}^c N_j * \log(N_j / N)}} \quad (11)$$

Where N: data size. c : Clusters number.

$N_i$ :Number of data points in clusters i.

$N_j$ :Number of data points in true cluster j.

$N_{ij}$ :Number of agreements between clusters i and true clusters j

The classification accuracy is the ratio of number of correct predictions by the data size multiplied by 100, its formula given by EQ.12

$$Accuracy = \frac{NCP}{N} * 100 \quad (12)$$

with : NCP:number of correct predictions

N:data size

The obtained clustering results are presented in Table.3 and Table.4.This tables compare the performance of our algorithm with the ESSC algorithm based on the NMI and classification accuracy.

The compared results demonstrate that our proposed algorithms exceeds the ESSC algorithms

From Tables 3 and 4, it can be seen that PSOESSC give a better results for data sets that have a big number of features and clusters (heart (18dim, wine(13 car 14), whilst the PSOALISSC showed a good results for database that contain a small number of clusters (iris balance and Australian).

Figure 1 presents the evolution of the PSOALISSC method. These curves confirm that the proposed method allows finding the optimum solution in a reasonable processing time.

TABLE I  
SETTING PSO PARAMETERS

|                   |          |
|-------------------|----------|
| Particles number  | [10 50]  |
| Iterations number | [10 150] |
| W                 | 0.8      |
| $c_1=c_2$         | 0.2      |

TABLE II  
UCI DATASETS[14]

| Dataset    | Clusters nbr | Instances nbr | Dimensions nbr |
|------------|--------------|---------------|----------------|
| Australian | 2            | 690           | 14             |
| Balance    | 3            | 625           | 4              |
| Car        | 4            | 946           | 18             |
| Glass      | 6            | 214           | 9              |
| Heart      | 2            | 270           | 13             |
| Iris       | 3            | 150           | 4              |
| wine       | 3            | 178           | 13             |

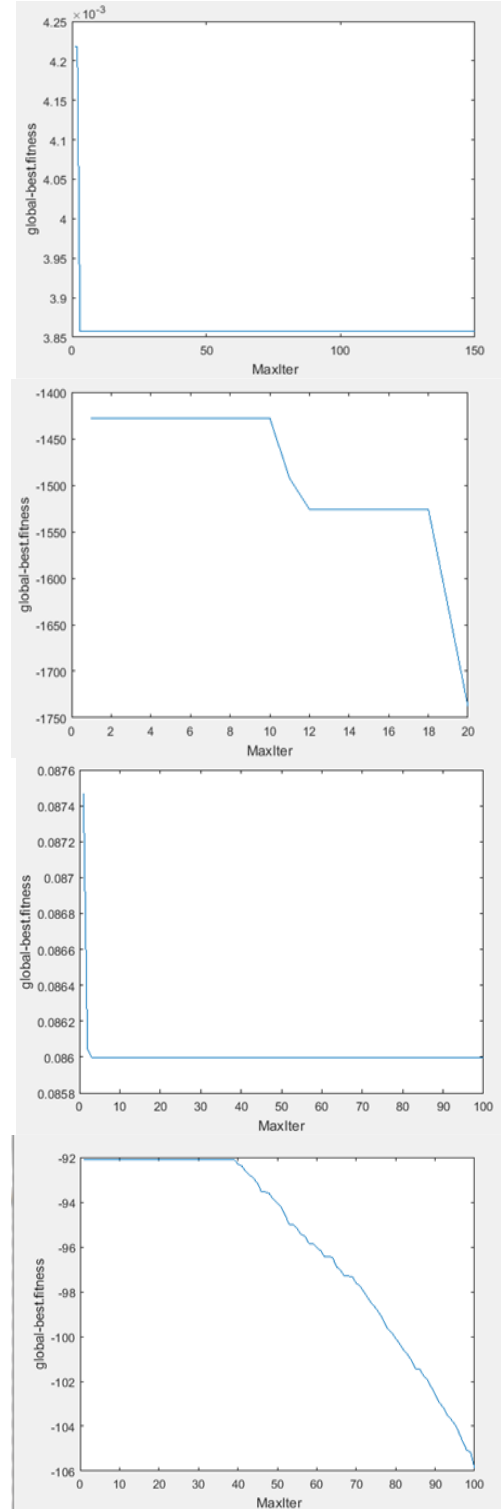


Fig. 1. Convergence of the PSOALISSC method to an optimal solution for the following bases (iris, balance, car, australian).

## B. Experiment on images

Synthetic and real images are tested to evaluate the performance of proposed algorithms.

Regarding aerial image classification, a preprocessing step is mandatory before applying the proposed methods; it consists in extract the subspaces or a features from the original image. The first three features were extracted by the co-occurrence matrix are: contrast, homogeneity and entropy. Grayscale and image edge detection based on Canny filter have been added. Therefore, we obtain five subspaces for each image which will construct the new data to be classified.

The clustering results of synthetic and Aerial images obtained by the proposed method and the ESSC algorithm are implemented in Figure.2 and Figure.3 respectively. We take the same number of classes for both methods.

The figures 2.c ,2.f and figure 2.i show. that all regions are well classified and well separated and that their contours are well limited, on the contrary to the results obtained by the ESSC algorithm where the two classes are merged presented by the (figures2.b and figure.2.e) The clustering results of Aerial images illustrated in figure.3.c and figure 3.f show that all regions (trees, sea, land, road) are well classified The trees are well detected and well separated from the other regions (Figure.3.c). The regions obtained by the proposed method (Figure.3.f) more homogeneous and sharper,the street edge are very clear and limited than the ESSC algorithm (figure (3.e)).

## IV. CONCLUSION

This paper, presents an clustering enhancement using particle swarm optimization. Soft subspace clustering and PSO are used alternately for have advantage of these both method. The

TABLE III  
NMI VALUES FOR UCI DATASETS

| Database   | PSOALISSC   |    | PSOESSC     |    | ESSC        |    |
|------------|-------------|----|-------------|----|-------------|----|
| Database   | NMI         | it | NMI         | it | NMI         | it |
| Australian | <b>0.96</b> | 5  | 0.39        | 14 | 0.36        | 12 |
| Balance    | <b>0.41</b> | 3  | 0.300       | 7  | 0.28        | 9  |
| Car        | 0.17        | 7  | <b>0.18</b> | 7  | 0.12        | 9  |
| Glass      | 0.61        | 12 | <b>0.76</b> | 9  | 0.35        | 12 |
| Heart      | 0.14        | 4  | 0.24        | 13 | <b>0.30</b> | 16 |
| Iris       | <b>0.86</b> | 5  | 0.86        | 5  | 0.74        | 16 |
| wine       | 0.75        | 9  | 0.80        | 11 | <b>0.86</b> | 12 |

TABLE IV  
CLASSIFICATION ACCURACY

| Dataset    | PSOALISSC    | PSOESSC      | ESSC  |
|------------|--------------|--------------|-------|
| Australian | <b>99.5</b>  | 88.55        | 80.65 |
| Balance    | <b>93.12</b> | 77.02        | 59.20 |
| Car        | <b>90.42</b> | 88.60        | 34.04 |
| Glass      | <b>71.49</b> | 57.00        | 41.50 |
| Heart      | 92.44        | <b>94.14</b> | 72.43 |
| Iris       | <b>98.67</b> | 98.66        | 90.00 |
| wine       | 81.57        | <b>88.20</b> | 45.37 |

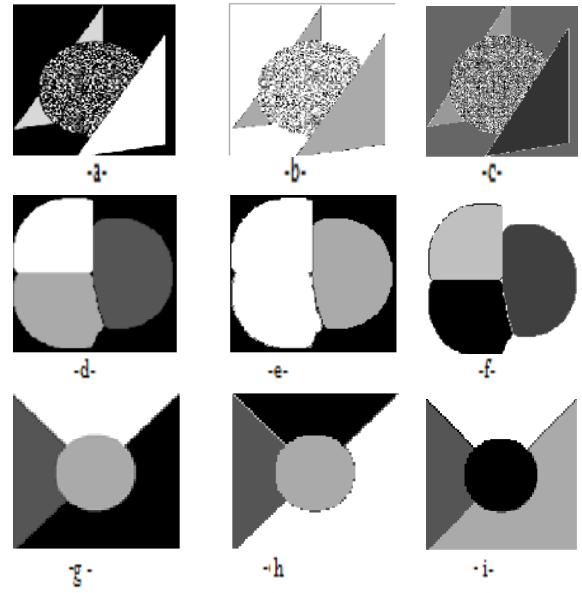


Fig. 2. clustering results.(a,d,g) : original images; (b,e,h) : ESSC algorithm, (c,f,i) :proposed approach

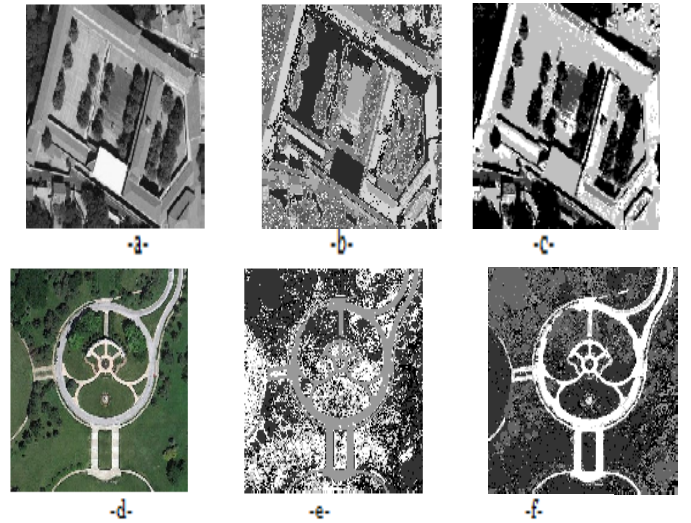


Fig. 3. (a,d) : original images; (b,e) : ESSC algorithm,(c,f) :proposed approach

PSO is able to find the most optimistic result, by modulating PSO parameters, convergence can be accelerated. The PSO offers significant improvements regardless of the objective function to be optimized. However, The quality of the objective function affects the quality of the clustering. PSO is very sensitive to the parameters used to obtain very good results.

Future work is about the the multi-objective clustering

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